SE3010 – Software Engineering Process & Quality Management
BSc (Hons) in Information Technology
Specialized in Software Engineering
3rd Year
Faculty of Computing
SLIIT

2026 – Assignment 02

Test Automation and Test-Driven Development

Course: SE3010 - Software Engineering Process and Quality Management
Weight: 20% of the final grade
Due Date: [18-04-2026]

Objective

This assignment aims to provide students with in-depth knowledge and application in software test automation and test-driven development (TDD), enabling them to understand their importance, benefits, and challenges in software development.

Instructions and Submission Guidelines

- This is a **group assignment** (4 members per group).
- Refer the lecture materials and other resources on test automation and TDD concepts and best practices.
- This assignment consists of **two parts**:

Part	Description	Marks
Part 1	Research Report	6/8 Marks
Part 2	Quiz (MCQs)	14/12 Marks
Total		**20 Marks**

- Plagiarism will not be tolerated; ensure you submit original work.
- Late submissions will result in deduction of marks.
- The following guidelines should be followed when submitting the report
  - **Format**: PDF Document.
  - **Naming**: Rename the document with the group ID
  - The names and registration numbers of all the members in the group should be included in the cover page of the report.
  - **Length**: 12-15 pages.
  - **Diagrams**: Include relevant visuals (flowcharts, graphs, etc.).

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PART 1 – Research Report (6 Marks)

Report Requirements

Students must prepare a **research-based report (minimum 15 pages)** covering **10 topics** related to **Test Automation and Test-Driven Development (TDD)**.

Report Structure

The report must include:

1. Introduction
2. Test Automation Topics (3 topics)
3. Test Driven Development Topics (3 topics)
4. Combined Test Automation & TDD Topics (4 topics)
5. Conclusion
6. References

Topics for the Report

A. Test Automation Topics

Students must research and write on:

1. Test Automation Lifecycle and Strategy
2. Test Automation Frameworks (Modular, Data-driven, Keyword-driven, Hybrid)
3. Integration of Test Automation with CI/CD and DevOps

B. Test Driven Development Topics

Students must research and write on:

4. TDD Lifecycle – Red, Green, Refactor
5. Benefits and Challenges of TDD in Real Projects

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6. SOLID Principles and Their Relationship with TDD

C. Combined Topics – Test Automation + TDD

Students must research and write on:

7. Using TDD with Automated Unit Testing Frameworks
8. Continuous Testing using TDD and Automation in CI/CD
9. Comparison: Traditional Testing vs TDD + Automation Approach
10. Role of Test Automation and TDD in Agile and DevOps Environments

Marking rubric for Part 1: Research Report

Criteria	High (6-8 Marks)	Average (3-5 Marks)	Low (0–2 Marks)
1. Report Topics			
1.1 Test Automation Lifecycle and Strategy	Clearly explains all lifecycle stages, automation strategy, tool selection, CI/CD integration, includes diagrams and real-world examples with discussion of challenges.	Explains lifecycle and strategy with some examples but limited depth or diagrams.	Basic or incomplete explanation with little understanding and no examples.
1.2 Test Automation Frameworks	Explains multiple frameworks (modular, data-driven, keyword, hybrid), compares them, includes diagrams and real-world usage scenarios.	Describes frameworks with some comparison but lacks examples or detailed analysis.	Lists frameworks with minimal explanation and no comparison.

1.3 Automation with CI/CD and DevOps	Clearly explains CI/CD pipelines, automation integration, continuous testing workflow, benefits and challenges with diagrams.	Explains CI/CD and automation integration but lacks workflow explanation or detailed discussion.	Basic explanation of CI/CD with limited understanding of automation integration.
1.4 TDD Lifecycle (Red–Green–Refactor)	Clearly explains Red-Green-Refactor cycle with examples, diagrams, and real-world scenarios including benefits.	Explains TDD cycle with examples but lacks diagrams or deeper discussion.	Basic explanation of TDD cycle without examples or clear understanding.
1.5 Benefits and Challenges of TDD	Discusses benefits and challenges with real-world examples and critical evaluation.	Lists benefits and challenges with some explanation but limited examples.	Only lists benefits or challenges with little explanation.
1.6 SOLID Principles and TDD	Explains all SOLID principles and clearly links them to TDD with examples.	Explains SOLID principles but limited discussion on relationship with TDD.	Basic explanation of SOLID principles without linking to TDD.
1.7 TDD with Automated Unit Testing	Clearly explains integration of TDD with automated unit testing tools, workflow, examples, and benefits.	Explains TDD and unit testing integration but lacks workflow or examples.	Basic explanation of unit testing or TDD separately.
1.8 Continuous Testing using TDD and Automation	Explains continuous testing, CI/CD integration, automation pipeline, and TDD role with diagrams/workflows.	Explains continuous testing and CI/CD but limited discussion on TDD integration.	Basic explanation of continuous testing only.
1.9 Traditional Testing vs TDD + Automation	Provides detailed comparison, advantages, disadvantages, diagrams, and critical evaluation of both approaches.	Provides comparison but limited evaluation or examples.	Basic description without comparison or analysis.

1.10 Agile/DevOps with TDD and Automation	Clearly explains Agile/DevOps integration, continuous testing, automation pipelines, TDD role, and real-world workflow.	Explains Agile/DevOps and TDD/Automation relationship but lacks workflow explanation.	Basic explanation of Agile or DevOps only with limited understanding.
2. Conclusions	**[8-10 Marks]** Strong conclusion summarizing findings, comparisons, recommendations, and future improvements	**[5-7 marks]** Good summary but limited critical discussion or recommendations.	**[0-4 marks]** Basic summary of report without analysis.
3. References	**[5 Marks]** Many academic references, proper citation style (IEEE/APA), credible sources.	**[3-4 Marks]** Some references but not properly formatted.	**[0-2 Marks]** Very few references.
4. Overall Structure and Communication	**[5 Marks]** Excellent structure, clear headings, logical flow, diagrams, tables, and proper formatting.	**[3-4 Marks]** Good structure but minor issues in organization or formatting.	**[0-2 Marks]** Poor structure, unclear organization.

Part 2 – Quiz Based on the Research Report

There will be 12/14 multiple choice questions prepared based on the topics from the research report (Part 1). This will be a lab test for 30 minutes where students will take the quiz on NetExam during lab hours.